

Algebra 1
Unit 6
Lesson 4 Practice

Name Answer Key
Date _____
Period _____

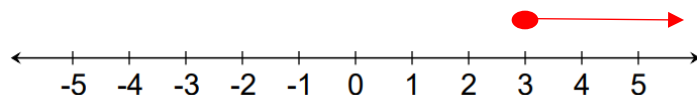
Consider the inequality $-3x + 4 \leq -5$.

1. Solve the inequality. $-3x + 4 \leq -5$
 $\quad\quad\quad -4 \quad -4$

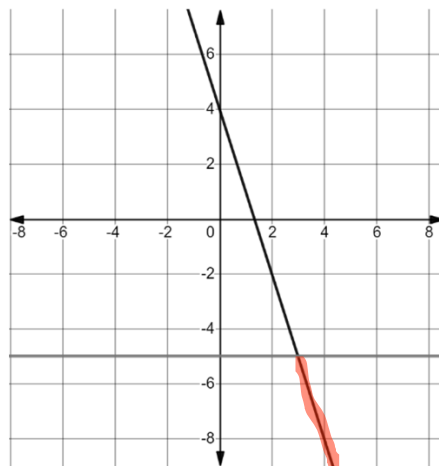
$$\frac{-3x}{-3} \leq \frac{-9}{-3}$$

$$x \geq 3$$

2. Graph the solution set.



3. The graphs of $y = -3x + 4$ and $y = -5$ are shown on the coordinate plane.

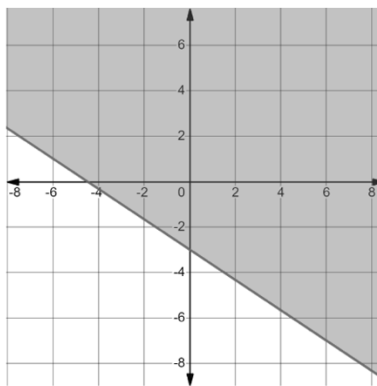


Highlight the part of the graph of $y = -3x + 4$, where the y values are less than or equal to -5 .

4. Describe the relationship between the solution to the inequality $-3x + 4 \leq -5$ and the values of x for which the graph of $y = -3x + 4$ is less than or equal to -5 .

When $x \geq 3$, the solution to $-3x + 4 \leq -5$, the y -values are less than or equal to -5 .

The graph of the two-variable inequality $y \geq -\frac{2}{3}x - 3$ is shown.



5. The slanted solid line is called the **boundary line** of the inequality. Write the equation of the boundary line in slope-intercept form.

$$m = \frac{-4}{6} = \frac{-2}{3} \quad y = \frac{-2}{3}x - 3$$

6. How does the equation of the boundary line relate to the inequality?

The equation of the boundary line is the same as the inequality except that it has an equal sign.

7. Describe what part(s) of the graph represent the solution set of the inequality $y \geq -\frac{2}{3}x - 3$.

All values on and above the boundary line are in the solution set to the inequality.

8. Which part of the graph of the solution set of $y \geq -\frac{2}{3}x - 3$ corresponds to each mathematical statement?

Mathematical Statement	Graphical Representation
$y > -\frac{2}{3}x - 3$	☐ boundary line ☒ shaded region
$y = -\frac{2}{3}x - 3$	☒ boundary line ☐ shaded region

Complete the statements.

9. If the points on the line are included in the solution set, the boundary line

is

- ☐ dotted.
☒ solid.

10. If the points on the line are not included in the solution set, the boundary line

is

- ☒ dotted.
☐ solid.